SMART GLASSES FOR PEOPLE WITH HOMONYMOUS HEMIANOPIA

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1.0 Executive Summary

Homonymous Hemianopia (H.H) is a visual impairment characterized by the loss of vision on the same side of both eyes, most often resulting from traumatic events such as stroke. Individuals

with H.H face increased safety risks due to limited environmental awareness. Despite this there are currently few solutions available to those with H.H especially none that do not introduce a new element of cognitive overload.

This project presents a new solution, a wearable smart glasses system that is designed to assist individuals with H.H by providing AI enabled object detection. This will in turn provide users with increased environmental awareness. An AI camera mounted to the glasses continuously captures the surrounding environment and identifies objects and potential hazards including people, animals, and common obstacles/objects. The item(s) that are detected with the highest confidence are then communicated to the user via short text labels displayed on a micro-OLED projector preserving the user's remaining vision by minimizing the intrusion.

The wearable prototype was built around a Raspberry Pi Zero 2 W, a Raspberry Pi AI camera, a micro-OLED display, beam splitters, and a custom 3D-printed set of frames. The entire system is powered by two Samsung 18650 lithium batteries via a UPS expansion board which provides 4 hours of continuous run time. A secondary bench model was also created which used 4 of the same batteries which has a run time of 8 hours at higher performance stereovision.

Testing confirmed that the adjustable display met its design requirement and live object identification produced positive results across more than 20 distinct objects. Testing occurred in both controlled and uncontrolled environments. The device was completed within its $10,000 budget at a unit cost of about $1,700. The primary unmet requirements are the 230g weight limit (current system weighs 413g) and the 8 hour battery life target (current system runs 4 hours). Both of these reflect the tradeoff of size vs use and should be the primary focus of any future research.

2.0 Problem Definition

2.1 Problem Scope

It is estimated that one third of all stroke survivors will experience some amount of vision loss. H.H is among the most common. People with H.H carry the risk of collision with obstacles, difficulty navigating crowded spaces, and potentially reduced independence in daily activities.

There are currently a very limited number of solutions. Optical prisms, one of the most frequently used tools, can shift a portion of the visual field from one area onto another. These require training and still have high rates of headaches and disorientation. Virtual reality has been explored but is currently impractical for daily wear. The result is that most people with HH simply change their personal behaviors by adopting frequent head scanning with no actual tools or support.

The scope of this project was to design, build, and test a wearable device the increases environmental awareness in the blind spaces of people with HH with minimal alteration of the user's remaining vision. This project also needed to be practical for extended daily use as measured by its weight, battery life, and easy of operation and maintenance.

2.2 Technical Review

There are three primary tools currently used for people with HH. These were reviewed prior to designing the product.

Prism Glasses:

These are the most commonly used tool for H.H. They displace a portion of the visual field from the lost portion into the seeing portion. This displaced display is partially see through allowing users to see two scenes in one area of vision. This does give users environmental awareness however also brings visual distortion in the prism zones. To be used at all users must train. If a user has any additional visual impairments then this solution becomes less helpful. No double-vision prism system will provide the awareness without the disorientation. The project sponsor, Dr. Tom Politzer has worked with patients with H.H and this route is commonly complained about and rejected.

Mirror Systems:

These are systems that use mirrors and/or camera to screen systems that relay the blind view onto a display. This display would take some of the space away from the portion of

their field that a person with H.H could see. If displaying a whole field back into the portion that a person with H.H can see then this solution runs into the same problem as the prisms of overloading the user.

AI Wearable Cameras:

Advances in low power AI modules have made them much more accessible and versatile. Currently low cost AI modules are being used for object detection in security and tracking purposes. This project will bring this technology into the medical sector.

2.3 Design Requirements

The design requirements established at project initiation by the project sponsor and agreed to by the team are listed in the following table, Table 2.1. The field of view requirement was reframed during the design and testing process, rather than expanding the patient's raw visual capacity by 60 degrees the final design uses camera placement and object detection to provide environmental awareness in the 60 degree region. This reframing was approved by the sponsor.

ID	Requirement	Target Value
1	System Weight	< 230g
2	Battery Life	>8 hours
3	Adjustable Display	Adjustable angle and focal distance
4	Budget	< $10,000
5	Coverage	60 degrees
6	Object Identification	Accurate and fast in live use

3.0 Design Description

3.1 Overview

The Pi Glasses are a wearable assistive device built on a part of custom 3D-printed PLA frames. The system integrates an AI camera, a Raspberry Pi Zero 2 W as the processor, a micro-OLED display, beam splitters, and two Samsung 18650 lithium batteries. The purpose of the device is to continuously capture the user's blind side environment identify the most significant objects and present this to the user with minimal interruption to their vision.

The wearable system is the primary deliverable of this project. A bench model incorporating dual stereovision cameras, four batteries, and a higher performance Raspberry Pi was developed as well to demonstrate the future of this project should the technology exist to make a minimized version.

3.2 Detailed Description

The wearable system has six subsystems:

Processing Unit:

A Raspberry Pi Zero 2 W serves as the onboard computer. This Pi was selected for its size, processing capabilities, and wireless connectivity (an essential part of this design). This unit is compatible with the Raspberry Pi AI Camera. An aluminum heatsink was attached to the processor to ensure longer runtimes with minimal damage and wear to the unit. The Pi Zero 2 W is mounted to a UPS expansion board along with a Samsung 18650 battery.

Camera Module:

A Raspberry Pi AI camera is mounted in a hinged rotating case which attaches to the glasses frames. The hinge allows for approximately 120 degrees of rotation enabling the camera to be angled as needed by the user. A set screw put in place by the user locks the camera at the desired angle. The camera captures live video which is processed by the Pi Zero 2 W running object detection through YOLO. The camera is powered by a second Samsung 18650 battery positioned on the opposite side of the frames for counter balance.

Optical Display System:

Once objects have been seen, detected, and confidence processed, these are communicated to the user via a micro-OLED display paired with a beam splitter and optical lens. The beam splitter is mounted on an adjustable hinge that positions is in the user's line of sight. The display projects the text labels from the Zero 2 W of the one or

two highest confidence detected objects directly into the beam splitter which reflects the image into the user's eye as a transparent overlay. This provides awareness with minimal intrusion into the user's vision. The display mount allows adjustment of both angular position and the focal distance of the projected image. Each individual will need unique combinations of aangualr position and focal distance so these being adjustable were essential to this product being able to be wide spread.

Frames:

The glasses frames were custom designed in OnShape and 3D printed in PLA. The frames incorporate modular mounting points for all electronic subsystems allowing components to be repostioned and attached using M2.5 fastner hardware (screws, washers, spacers, and nuts). The frames were structurally analyzed using SolidWorks FEA simulations prior to fabircatio to verify that it could support the weight of the components under normal day to day use.

Power System:

The wearable system is powered by two Samsung 18650 lithium ion cells. One is installed in the UPS expansion board while the other site ontop the frames opposite the camera. At current component power draw there is an estimated 4 hours of continuous operation. The UPS board includes a USB charging port for recharging.

Fasteners:

M2.5 x 6+6 brass threaded spacers, M2.5 washers, M2.5 nuts, and PM M2.5 x 6 screws (30 each) are used throughout the assembly to mount all subsytems to the frames and to each other.

3.3 Use

The Pi Glasses are intended for daily use by individuals with low vision or visual field deficits, particularly those with Homonymous Hemianopia, who would benefit from enhanced awareness of objects in their blind-side environment. The device is worn like a standard pair of glasses. Once powered on, the system operates autonomously: no user interaction is required during normal operation. The AI detection model continuously processes the camera feed, and when objects are identified above the confidence threshold, their names are displayed as short text labels on the micro-OLED overlay.

Users should complete the calibration procedure (detailed in the Technical Instructions) prior to first use to align the display to their individual focal geometry and to orient the camera toward the intended coverage area. The device is not intended as a substitute for clinical low-vision rehabilitation and should be used in conjunction with guidance from a qualified vision care professional.

The device provides approximately four hours of operation per charge on the wearable prototype. Users should charge the device daily and should not use it as a primary safety device in high-risk environments (e.g., driving) until further clinical validation has been completed. Refer to the Technical Instructions document for detailed assembly, calibration, maintenance, and troubleshooting procedures.

4.0 Design Evaluation

4.1 Overview

4.1.1 Test Plan

The Pi Glasses system was evaluated through a combination of physical prototype testing, theoretical calculations, and virtual structural simulation. Physical measurements were used for weight verification. Theoretical power calculations were used to estimate battery life, chosen over direct continuous-operation testing to protect the micro-OLED display from potential damage during unsupervised extended-duration runs. SolidWorks finite element analysis (FEA) was used for structural evaluation of the 3D-printed PLA frames. Live object detection testing was performed to assess the AI identification capability of the system in both controlled and real-world conditions.

The design requirements are listed in Section 2.3 and are used for the evaluations. The summary of results is listed below.

Table 1: Summary of Results

ID	Requirement	Target Value	Test Method	Result
1	Weight Limit	< 230g	Scale Measurement	FAIL – 413g
2	Battery Life	>8 hours	Theoretical: Capacity/Draw	FAIL – 4 hours
3	Adjustable Display	Functional Adjustment	Physical Verification	PASS
4	Budget	< $10,000	Cost Tracking	PASS - $1,700
5	Coverage	60 degrees	Camera Angle Tracking	PASS
6	Object Identification	Hazards in real time	Live testing	PASS

4.2 System and Components

4.2.1 System Weight

Purpose: The weight of the wearable device was evaluated to verify that the system met the design requirement of 230 grams or less. Staying within this limit is critical to user comfort and long-term wearability, particularly for a patient population that may wear the device for extended periods throughout the day.

Test Methods: Each component of the Pi Glasses system was weighed individually using a calibrated digital scale. The individual weights were then summed to determine the total system weight. Weighing components individually allowed identification of any single contributor that disproportionately affected the total, enabling targeted weight reduction decisions.

Results and Discussion: The total system weight measured 413 grams, exceeding the 230-gram requirement by 183 grams. The Pi configuration — comprising the Raspberry Pi Zero 2 W, heatsink, UPS expansion board, single battery, and casing — is the largest single contributor to total weight. Within that assembly, the 18650 battery cell accounts for the greatest single-component mass at 63 grams. This requirement was not met. Importantly, the most straightforward path to meeting the battery life requirement (adding a second battery in parallel) would increase total system weight further, reinforcing the fundamental tradeoff between battery capacity and weight compliance described in Section 4.3.

4.2.2 Battery Life

Purpose: The battery life of the wearable system was evaluated to determine whether the device could sustain eight hours of continuous operation per charge, as specified in the design requirements. Adequate battery life is essential for the device to be practical for full-day daily use.

Test Methods: Battery life was evaluated using a theoretical calculation rather than direct continuous-operation testing. This approach was chosen to protect the micro-OLED display from damage that could result from unmonitored extended-duration operation of the early prototype. Battery life was estimated using the standard relationship: Runtime (hrs) = Battery Capacity (mAh) / Total System Power Draw (mA). Power draw values for each active component were sourced from manufacturer datasheets and summed to obtain a total system draw estimate.

Results and Discussion: The wearable prototype, operating from a single 18650 battery cell, was estimated to provide approximately four hours of continuous runtime, falling short of the eight-hour target. Adding a second battery in parallel to the UPS board would theoretically double runtime to approximately eight hours; however, this would add approximately 63 grams to the total system mass, pushing the device further above the 230-gram weight limit. The bench model, which houses four batteries and a higher-performance processor, exceeds the eight-hour battery life target and serves as the performance benchmark for the next-generation wearable design.

4.2.3 Adjustable Display

Purpose: The adjustable display requirement was evaluated to confirm that the micro-OLED projection system could be repositioned to accommodate the unique focal geometry of individual users. Because each user's eye has a distinct focal point and preferred viewing angle, adjustability enables a single manufactured design to be fitted to a wide range of patients without custom fabrication.

Test Methods: The display mount was physically adjusted across its full range of motion, varying both the angle of the beam splitter and the distance of the micro-OLED from the user's eye. Multiple team members with different visual geometries wore the device and confirmed whether the projected image could be positioned to be clearly visible and comfortably located within the line of sight without obstructing residual vision.

Results and Discussion: The adjustable display mechanism functioned as designed and met the requirement. The mount provided sufficient range of angular and distance adjustment to accommodate user differences in focal geometry. This result is significant for manufacturability: by designing for adjustability rather than custom fitting, the system remains practical to produce at scale while still serving a patient population with inherently variable optical needs.

4.2.4 Frame Structural Integrity

Purpose: A structural analysis of the 3D-printed PLA glasses frames was conducted to verify that the frames could withstand the mechanical loads associated with normal use and handling without failure. The frames are the primary load-bearing component of the system and support all electronic hardware.

Test Methods: A CAD model of the frames was imported into SolidWorks and analyzed using the SolidWorks Simulation FEA tool. The frame geometry was

meshed and a static load case representing standard standing wear was applied: a distributed load across the top of the frames with point load concentrations at component mounting locations, and fixed constraints applied at the contact surfaces between the frames and the user's face. Material properties corresponding to PLA (Young's modulus ≈21 GPa, tensile strength ≈58 MPa) were applied.

Results and Discussion: The load cases were within the acceptable stress and deformation limits and passed the structural evaluation.

4.2.5 Object Identification and Environmental Awareness

Purpose: The AI-enabled object identification system was evaluated to determine whether the camera and processing hardware could reliably detect and classify objects and hazards in the user's environment. This is the core functional requirement of the device: the system's clinical value depends entirely on accurate and timely identification of objects in the affected visual field.

Test Methods: The object identification system was tested by presenting a variety of objects to the camera in both controlled single-object scenarios and in a populated room environment. Objects were selected to represent the hazard categories most relevant to an H.H. patient navigating daily environments: people, animals, furniture, and common household and outdoor obstacles. The system's micro-OLED output was observed for each object and classified as a correct identification, a misidentification, or a failure to detect. Testing in more complex dynamic environments and evaluation of distance estimation performance were identified as open items requiring further evaluation.

Results and Discussion: The object identification system produced positive results across more than 20 distinct object types, including testing performed in a moderately crowded room. The system demonstrated a consistent ability to detect and classify objects in real-world conditions beyond a simple bench test. These preliminary results are encouraging for the intended use case, as patients will encounter the device in everyday, dynamic environments rather than controlled settings. Quantitative metrics including average response latency and false positive rate are planned for the next phase of evaluation. Distance estimation and performance in densely populated or fast-moving scenes remain as open test items.

4.3 Conclusions

The Pi Glasses system for individuals with Homonymous Hemianopia demonstrated meaningful progress toward meeting the stated design requirements, with notable successes in adjustable display functionality, budget compliance, environmental coverage by design, and preliminary object identification capability.

The adjustable display mount performed as designed, enabling a single manufactured unit to be fitted to users with varying focal geometries. This is a clinically meaningful result: it demonstrates that the device does not require custom fabrication for each patient, which is a prerequisite for any practical clinical deployment or future commercialization. The project was completed at a cost of approximately $1,700, well within the $10,000 budget, leaving substantial margin for component upgrades and iterative development.

The most significant unmet requirements at project close are the 230-gram weight limit and the 8-hour battery life target. At 413 grams, the current wearable prototype is approximately 80% heavier than the requirement. The four-hour estimated battery runtime falls half a day short of the target. These two failures are not independent: they arise from the same root cause. The 18650 battery cell and its associated power management hardware are the dominant weight contributors, and any straightforward increase in battery capacity (adding cells in parallel) increases weight proportionally. This is the critical engineering tradeoff remaining in the design.

The bench model demonstrates that the performance targets are achievable: four batteries provide sufficient runtime, dual cameras enable stereovision and distance estimation, and a more powerful processor handles denser detection scenes. The development path is therefore not a redesign from first principles but a miniaturization problem — reducing the bench model architecture to wearable dimensions and mass. Until the weight and battery life requirements are met and object identification testing is completed with quantitative metrics, the device is not ready for patient use.

Overall, the Pi Glasses system is a strong proof-of-concept. The core technical approach — using AI object detection to selectively communicate environmental hazards via a minimal heads-up display, without obstructing remaining vision — has been demonstrated to function in real-world conditions. The design choices made to prioritize minimal visual intrusion and adjustability are clinically well-founded. The path to a clinically deployable device is clear; it requires engineering refinement rather than architectural change.

4.4 Next Steps

The primary next step is to miniaturize the bench model architecture into a wearable form factor. The bench model currently outperforms the wearable prototype in all key metrics: four battery packs provide extended runtime, dual cameras with stereovision enable distance estimation, and a higher-performance processor handles denser object identification scenes. The development objective is to reduce the physical footprint and mass of these components to wearable dimensions while retaining their performance advantages.

Before the device can be considered ready for patient use, the following evaluations must be completed:

- Battery and weight reconciliation: Final power draw measurements should be taken from the assembled wearable system under operational conditions and compared against the theoretical estimates. If a two-battery configuration is pursued to meet the eight-hour target, total system weight must be re-measured and evaluated against the 230-gram limit. If weight remains non-compliant, frame material alternatives (e.g., PETG, lightweight composites) and component consolidation strategies should be explored.
- Object identification quantitative testing: Detection accuracy rate, average response latency, and false positive rate should be recorded across a standardized object set. Testing in highly dynamic and cluttered environments should be completed. Distance estimation accuracy using the bench model stereovision system should be characterized and a target accuracy threshold established.
- User testing: Once the device is mechanically and functionally stable, formal user testing with individuals who have Homonymous Hemianopia, or clinical proxies established in consultation with Dr. Politzer, should be conducted to assess usability, comfort, and real-world effectiveness.
- SolidWorks FEA values: Final numerical stress and deformation values from the frame structural analysis should be recorded and inserted into Table 4.2 of this report.

The team taking this project forward should begin by pursuing an alternative power strategy: higher energy density battery chemistry (e.g., lithium polymer cells), dynamic power throttling during low-activity periods, or component consolidation to reduce the overall component count and mass. Object identification testing in more demanding environments should follow. The device should not be transitioned to patient use until all design requirements have been formally verified and any safety-related failure modes have been assessed and mitigated.

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